Nikita Kolosov

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EDUCATION

The Ohio State University - PhD candidate, Biomedical Informatics — 2023-present
Biomedical Sciences Graduate Program • Post-candidacy — Expected Graduation: May 2028
ITMO University - MSc, Bioinformatics and Systems Biology — 2020
Saint Petersburg State University - BSc, Developmental Biology — 2018

EXPERIENCE

The Ohio State University / Nationwide Children's Hospital - *Graduate Research Associate* — 2023-present

- Developed **PGS Browser**, a platform for polygenic risk prediction and interpretation with a web interface, backend, and Dockerized CLI; Integrated it into internal FinnGen Sandbox infrastructure; the browser is **actively used by over 1,000 researchers**. First-author publication in *Nature Communications* (2026). Resource: www.pgs.nchigm.org.
- Evaluated **3,168 polygenic score models** across **4,739 disease endpoints** in **473,681 FinnGen participants** and released population reference distributions to contextualize individual scores.
- Developed prediction models integrating genetic and clinical risk factors using regularized logistic and Cox regression to estimate disease risk and time to disease onset.
- Developed **Nextflow pipelines** to analyze pediatric whole-genome sequencing (WGS) data from trios and duos, identify candidate variants, and characterize pharmacogenomic (PGx) variation.

Broad Institute of MIT and Harvard & FinnGen - *Associated Researcher.* — 2020-present

- Developed **WDL pipelines and reproducible HPC workflows** for biobank-scale genotype processing, polygenic risk modeling, and metabolomic, proteomic, and RNA-seq analyses, contributing to publications in *Nature Genetics*, *Nature Communications*, and *AJHG*.
- Developed **GPrior**, a **disease-gene prioritization** tool integrating **positive-unlabeled learning**, functional annotations, and expert priors; first-author publication in *European Journal of Human Genetics* (2021). Code: www.github.com/faramer86/GPrior.

Earlier roles: Research Consultant, MGH (2022-2023); Staff Scientist, Almazov Centre (2020-2022); Junior Programmer, ITMO University (2019-2022); Junior Bioinformatician, Parseq Lab (2019-2020).

SELECTED PUBLICATIONS

Kolosov N, et al. PGS Browser: a public platform for personalized polygenic score analysis and interpretation. *Nature Communications* 17, 7806 (2026). doi:10.1038/s41467-026-74461-7

Usoltsev D, **Kolosov N**, et al. Complex trait susceptibilities and population diversity in a sample of 4,145 Russians. *Nature Communications* 15, 6212 (2024). doi:10.1038/s41467-024-50304-1

Kolosov N., Daly M.J. and Artomov M. Prioritization of disease genes from GWAS using ensemble-based positive-unlabeled learning. *European Journal of Human Genetics* (2021). doi:10.1038/s41431-021-00930-w

Additional coauthored research: autoimmune hypothyroidism, *Nature Genetics* (2026); lichen planus, *AGHG* (2024).

Full list of publications: https://tinyurl.com/3chmfcth

TECHNICAL SKILLS

Programming and software development: Python, R, Bash/Linux, Git, Docker; Shiny, Dash.

Workflows and computing: Nextflow, WDL, Slurm, UGER, AWS S3.

Machine learning and statistics: Supervised/Unsupervised learning, deep learning, survival analysis, causal inference, Bayesian statistics, positive-unlabeled learning; scikit-learn, scikit-survival, PyTorch, glmnet, caret, XGBoost**.**

Statistical genetics and genomics: GWAS, QTL analysis, Mendelian randomization, PGS/PRS, PheWAS, whole-genome sequencing, trio/duo analysis, variant calling, imputation, pharmacogenomics; FinnGen, UK Biobank.

Multi-omics: Metabolomics (Nightingale), proteomics (Olink, SomaLogic), transcriptomics (RNA-seq).

Coursework: Statistical Machine Learning, Statistical Genetic Analysis, Bayesian Analysis, Regression Analysis, Survival Analysis, Causal Inference, Linear Algebra.

AWARDS AND RESEARCH COMMUNICATION

ASHG Trainee Research Excellence Finalist Award (2024); Outstanding Graduate Student Award, Abigail Wexner Research Institute (2024); ASHG platform talk (2024) and poster (2023); JetBrains research stipend (2020-2021); Winner of evolutionary bioinformatics hackathon (2018).